

Brent Machado

Software Engineer | Castlegar, British Columbia | [linkedin.com/in/brent-machado](https://www.linkedin.com/in/brent-machado) | [bmachado18.github.io](https://github.com/bmachado18) | brentmachado7@gmail.com | 2506082440

Innovative and dynamic developer with hands-on experience in backend design, end-to-end testing, machine learning, and cloud-based workflows.

Skills

Python Java Linux HTML, CSS, Javascript NGINX Docker AWS SQL
NodeJS OpenCV Computer Vision

Work Experience

Selkirk Innovates Graduate Researcher

Sep 2024 - Current

- Redesigning FastAPI backend to scale, leveraging Celery and Redis for multitasking
- Lead project management to ensure application delivery
- Deployed isolated VMs for distributed computing on Digital Oceans
- Managed Ubuntu Linux servers, leveraging tools such as systemd, journalctl, ssh, and cron
- Created high quality and concise documentation to support knowledge sharing and maintenance

Hummingbird Drones Software Developer

Sep 2024 - Dec 2024

- Built end-to-end testing pipelines with Playwright for React
- Automated deployment workflows using AWS CodePipeline to avoid costly manual testing
- Trained hotspot detection models on AWS EC2 instances for scaling with ease
- Worked with cross-functional teams, demonstrating adaptability and strong communication skills

Selkirk Innovates Research Intern

May 2023 - Aug 2023

- Wrote Python and Bash scripts for data wrangling, processing, and automation
- Applied version control standards and data version control with Git and DVC
- Deeply analyzed academic papers for object detection methods with traditional and modern computer vision techniques
- Trained models on the DRAC high-performance clusters, leveraging SLURM
- Created and maintained high quality documentation to support team collaboration and knowledge sharing

University of Victoria Business Analyst

Jan 2022 - Aug 2022

- Automated mail preprocessing with Python and OCR technology to improve efficiency in data entry.
- Used unsupervised learning to segment donors, creating insights via Tableau dashboards
- Built email templates, surveys, and web pages using HTML, CSS, and JavaScript
- Queried and managed data efficiently using SQL for reporting and analysis for Microsoft SQL Server
- Collaborated with stakeholders to deliver tailored, data-driven solutions.

Education

University of Victoria Bachelors of Software Engineering

Sep 2020 - Apr 2025

- With a specialization in Artificial Intelligence and Machine Learning

Selkirk College Associate of Sciences

Sep 2018 - Apr 2020

- Focused on chemistry and completed with honors